Umm Al-Qura University
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SmartMajor

A Shell Script Project for Predicting Suitable University Majors

- Team Members:

Name	ID
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- Task Distribution:

Name	Tasks
Asail Hamdan	smartmajor_predict.sh
Danah Alharbi	advisor_panel.sh
Jomanah Alghamdi	smartmajor_stats.sh
Hajer Sayah	search_tool.sh & protect_data.sh

Supervised by: Dr. Eman Alharbi

1. Introduction

1.1 Problem Statement

Choosing a university major is a significant step in every student's academic journey. It has a direct impact on their future studies and career opportunities. However, many students face difficulties when making this decision, especially if they are not fully aware of what each major involves.

Some students choose a major based on social pressure, family influence, or their friends' choices, rather than selecting what truly matches their interests and abilities. This can lead to low motivation, academic struggles, and frequent changes in majors. In some cases, students may even leave university.

To help with this issue, we developed a simple tool that guides students in choosing the most suitable major. The system asks a series of yes/no questions about the student's interests and compares the answers with a dataset of majors. It then recommends a major that fits their personal preferences. Our goal is to make the decision process easier, clearer, and more personal.

1.2 Project Overview

SmartMajor is a smart system built on scripts that helps analyze students' interests and helps them choose the appropriate university major based on a student dataset. The student answers a set of yes or no questions, and the results are then compared with the Interests_Dataset.csv file, which contains a list of majors, and the interests related to each major.

System scripts:

- Predicting the appropriate major: **smartmajor_predict.sh**
- Viewing results and statistics: **smartmajor_stats.sh**
- Academic advisor control panel: **advisor_panel.sh**
- Search tool to access results by student name, university major, or date: **search_tool.sh**
- Data protection: **protect_data.sh**

1.3 Motivation

The motivation behind the Smart Major project is to help students make this important decision without distraction and confusion. We were in their place one day. Instead of being influenced by social pressures or random choices, students can predict the most appropriate major based on their interests in a scientific and technical manner. The use of Bash Script also adds a strong technical aspect and demonstrates how simple tools can make a difference and have a powerful impact on education and career guidance.

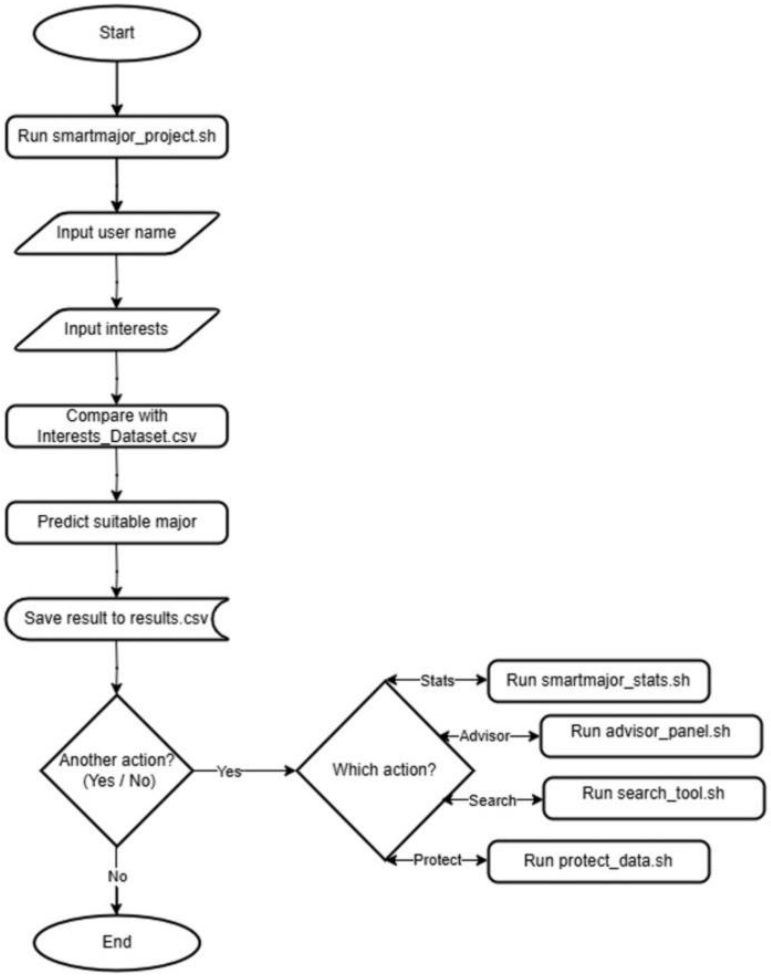
2. Design and Implementation

2.1 Detailed explanation of the chosen approach

The project predicts the best university major based on user preferences using a modular Bash scripting technique. A CSV dataset uses binary values to associate majors with particular interests. The script determines a matching score by comparing the user's yes/no answers to questions about their interests with the profiles of each major. It is advised to choose the major with the greatest score. The system is well-structured and user-friendly from the command line thanks to additional scripts that manage result viewing, analytics, searching, and data security.

2.2 Diagram

Flowchart of the SmartMajor system showing the steps from predicting a suitable major to accessing extra features such as statistics, search, advisor panel, and data protection.



2.3 GitHub Usage and Bash Script Deployment

We used GitHub for version control and collaboration. All code and dataset files were pushed from the terminal using standard Git commands.

- **GitHub Repository:** **https://github.com/asail-12/Software-Tools-Project.git**
- **Dataset Used:** **Interests_Dataset.csv** **:**

1	major	programming	problem_solving	biology	drawing	debate	leadership	technology	healthcare	math	reading
2	Computer Science	1	1	0	0	0	1	1	0	1	0
3	Medicine	0	0	1	0	0	0	0	1	0	1
4	Engineering	1	1	0	1	0	0	1	0	1	0
5	Business Administration	0	1	0	0	0	1	0	0	1	1
6	Media and Communication	0	1	1	0	0	0	0	0	0	0
7	Design And Arts	0	1	0	0	0	0	0	0	0	0
8	Social Work	0	0	1	0	1	0	0	0	0	0
9	Law	0	0	0	0	1	1	0	0	0	1

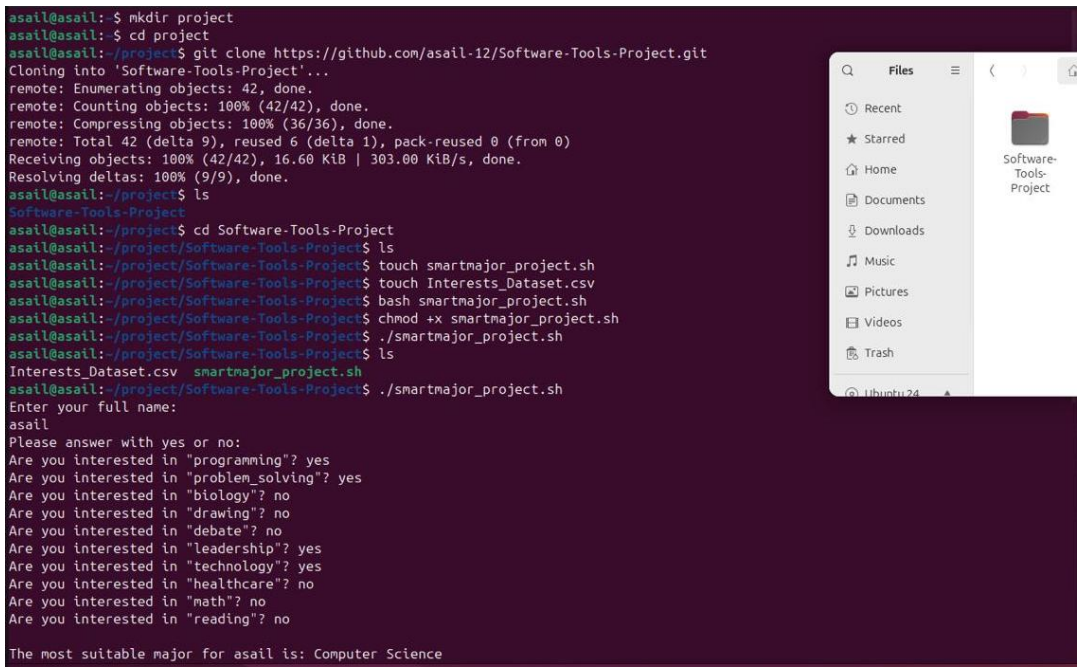
- The following picture shows the Git commit history for the uploaded Bash scripts and contributions from different team members.

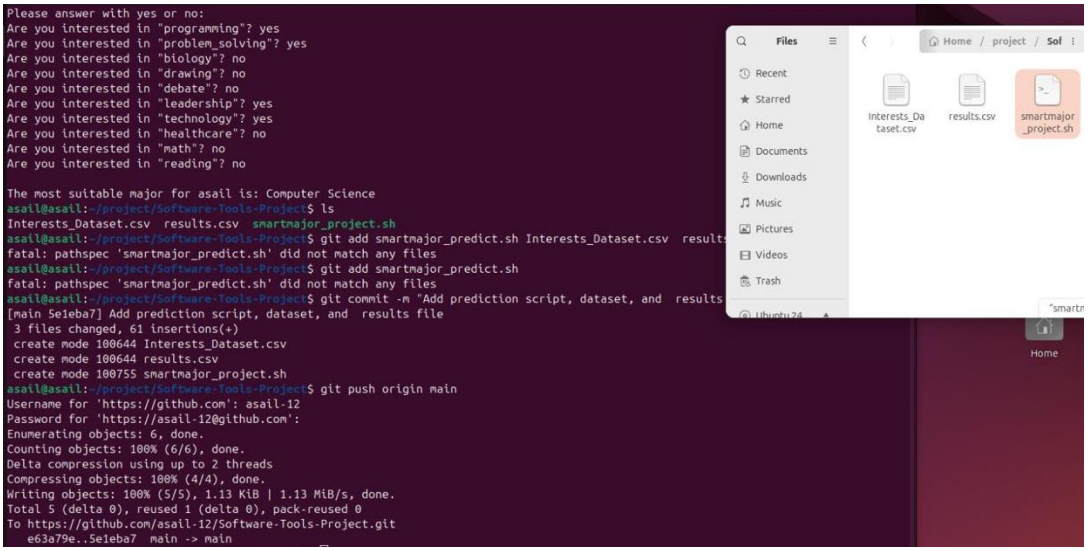
2.4 Bash Scripts: Upload, Source Code, and Output Demonstration

2.4.1 smartmajor_predict.sh

This script asks the user about their interests and predicts the most suitable university major based on their answers, using the Interests_Dataset.csv file for comparison.

- **Terminal Upload Commands**





- Script Code

```bash
#!/bin/bash

# Set dataset and results file names
dataset="Interests_Dataset.csv"
results_file="results.csv"

# Check if dataset file exists
if [ ! -e "$dataset" ]; then
  echo "Dataset $dataset not found!"
  exit 1
fi

# Ask for user's full name
echo "Enter your full name:"
read name

# Read the first line (headers) from the dataset
first_line=$(head -n 1 "$dataset")
IFS=',' read -r -a all_columns <<< "$first_line"

# Extract interest columns (excluding the first column)
interests=("${all_columns[@]:1}")

# Collect user's answers
answers=()
echo "Please answer with yes or no:"
for interest in "${interests[@]}"; do
  echo -n "Are you interested in \"$interest\"? "
  read response
  if [[ "$response" == "yes" || "$response" == "y" ]]; then
    answers+=(1)
  elif [[ "$response" == "no" || "$response" == "n" ]]; then
    answers+=(0)
  else
    answers+=(0)
  fi
done

# Initialize best match variables
best_major=""
max_score=0

# Compare user's answers with each major's interest profile
while IFS=',' read -r -a row_array; do
  if [[ "${row_array[0]}" == "major" ]]; then
    continue
  fi

  score=0
  for ((i=1; i<${#row_array[@]}; i++)); do
    if [[ "${answers[i-1]}" -eq 1 && "${row_array[i]}" -eq 1 ]]; then
      ((score++))
    fi
  done

  if (( score > max_score )); then
    max_score=$score
    best_major="${row_array[0]}"
  fi
done < "$dataset"

# Show and save the best matching major
echo ""
echo "The most suitable major for $name is: $best_major"
echo "$name,$(date +%F),$best_major" >> "$results_file"
```

- **The Output**

```
asail@asail:~/project/Software-Tools-Project$ ./smartmajor_project.sh
Enter your full name:
asail
Please answer with yes or no:
Are you interested in "programming"? yes
Are you interested in "problem_solving"? yes
Are you interested in "biology"? no
Are you interested in "drawing"? no
Are you interested in "debate"? no
Are you interested in "leadership"? yes
Are you interested in "technology"? yes
Are you interested in "healthcare"? no
Are you interested in "math"? no
Are you interested in "reading"? no

The most suitable major for asail is: Computer Science
```

- The output shows the analysis result based on the students' interests. After answering a set of questions, the system compares the answers against the dataset. In this example, "Computer Science" was selected as the most suitable major. The result is automatically recorded in the results.csv file.

2.4.2 smartmajor_stats.sh

This script analyzes results.csv to show how many students chose each major and provides general statistics.

- **Terminal Upload Commands**

```
vboxuser@Jomanah:~$ mkdir project
vboxuser@Jomanah:~$ cd project
vboxuser@Jomanah:~/project$ git clone https://github.com/asail-12/Software-Tools-Project.git
Cloning into 'Software-Tools-Project'...
remote: Enumerating objects: 45, done.
remote: Counting objects: 100% (45/45), done.
remote: Compressing objects: 100% (38/38), done.
remote: Total 45 (delta 9), reused 9 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (45/45), 16.95 KiB | 289.00 KiB/s, done.
Resolving deltas: 100% (9/9), done.
vboxuser@Jomanah:~/project$ ls
Software-Tools-Project
vboxuser@Jomanah:~/project$ cd Software-Tools-Project
vboxuser@Jomanah:~/project/Software-Tools-Project$ ls
Interests_Dataset.csv  results.csv  smartmajor_project.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ git add smartmajor_stats.sh
fatal: pathspec 'smartmajor_stats.sh' did not match any files
vboxuser@Jomanah:~/project/Software-Tools-Project$ git add smartmajor_stats.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   smartmajor_stats.sh

vboxuser@Jomanah:~/project/Software-Tools-Project$ git commit -m "Added smartmajor_stats.sh script"
[main 0487e64] Added smartmajor_stats.sh script
 1 file changed, 35 insertions(+)
 create mode 100755 smartmajor_stats.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ git config --global user.name "JomanahAhmeds"
vboxuser@Jomanah:~/project/Software-Tools-Project$ git config --global user.email "jomanahahmeds@gmail.com"
vboxuser@Jomanah:~/project/Software-Tools-Project$ git commit -m "Another commit just in case"
```

```
vboxuser@Jomanah:~/project/Software-Tools-Project$ git config --global user.name "JomanahAhmeds"
vboxuser@Jomanah:~/project/Software-Tools-Project$ git config --global user.email "jomanahahmeds@gmail.com"
vboxuser@Jomanah:~/project/Software-Tools-Project$ git commit -m "Another commit just in case"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
vboxuser@Jomanah:~/project/Software-Tools-Project$ git push origin main
Username for 'https://github.com': JomanahAhmeds
Password for 'https://JomanahAhmeds@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 812 bytes | 67.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/esail-12/Software-Tools-Project.git
   5e1eba7..0487e64  main -> main
```

- Script Code

```bash
#!/bin/bash

# Set the name of the results file
results_file="results.csv"

# Check if the file exists
if [[ ! -f $results_file ]]; then
  echo "No results found. Please run the prediction script first"  # Show message if the file is missing
else
  # Show the menu to the user
  echo "View and Statistics Menu:"
  echo "1. View all student results"
  echo "2. View statistics summary"

  # Ask the user to enter a choice
  read -p "Enter your choice (1 or 2): " choice

  # If the user chooses 1, show all results
  if [[ "$choice" == "1" ]]; then
    echo "Student results:"
    cut -d',' -f1,2,3 "$results_file"  # Display name, ID, and major

  # If the user chooses 2, show summary statistics
  elif [[ "$choice" == "2" ]]; then
    total=$(wc -l < "$results_file")  # Count how many students (lines)
    echo "Total students: $total"
    echo "Major predictions"
    # Show how many students got each predicted major (from most to least)
    cut -d',' -f3 "$results_file" | sort | uniq -c | sort -nr

  # If the input is not (1 or 2)
  else
    echo "Invalid choice. Please enter either 1 or 2"
  fi
fi
```

- **The Output**

```
vboxuser@Jomanah:~/project/Software-Tools-Project$ chmod +x smartmajor_stats.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ ./smartmajor_stats.sh
View and Statistics Menu:
1. View all student results
2. View statistics summary
Enter your choice (1 or 2): 1
Student results:
asell,2025-05-17,Computer Science
vboxuser@Jomanah:~/project/Software-Tools-Project$ ./smartmajor_stats.sh
View and Statistics Menu:
1. View all student results
2. View statistics summary
Enter your choice (1 or 2): 2
Total students: 1
Major predictions
      1 Computer Science
```

- The output shows how many students got each major based on the results.csv file. This helps us see which majors are more common and which ones are less chosen.

2.4.3 advisor_panel.sh

This script displays all student records from the results.csv file and shows basic statistics, including the most and least selected majors.

GitHub Push Issue

While uploading the advisor_panel.sh script to GitHub, I encountered a permission error. The terminal returned a 403 – permission denied message, which means my GitHub account did not have access to push changes to the repository.
This prevented the script from being uploaded directly by me and required access changes or help from a team member to complete the upload.

- **Terminal error during during push attempt for advisor_panel.sh**

```
ubuntu@ubuntu:~/Desktop/project-Test/Test$ git push origin main
Username for 'https://github.com': danah-ai
Password for 'https://danah-ai@github.com':
remote: Permission to danah-ai/Test.git denied to danah-ai.
fatal: unable to access 'https://github.com/danah-ai/Test.git/': The requested U
RL returned error: 403
```

- Terminal Upload Commands

```
vboxuser@Jomanah:~$ mkdir project
vboxuser@Jomanah:~$ cd project
vboxuser@Jomanah:~/project$ git clone https://github.com/asail-12/Software-Tools-Project.git
Cloning into 'Software-Tools-Project'...
remote: Enumerating objects: 80, done.
remote: Counting objects: 100% (80/80), done.
remote: Compressing objects: 100% (64/64), done.
remote: Total 80 (delta 28), reused 34 (delta 10), pack-reused 0 (from 0)
Receiving objects: 100% (80/80), 23.03 KiB | 491.00 KiB/s, done.
Resolving deltas: 100% (28/28), done.
vboxuser@Jomanah:~/project$ ls
Software-Tools-Project
vboxuser@Jomanah:~/project$ cd Software-Tools-Project
vboxuser@Jomanah:~/project/Software-Tools-Project$ ls
advisor_panel.sh       protect_data.sh  search_tool.sh         smartmajor_stats.sh
Interests_Dataset.csv  results.csv      smartmajor_project.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ git add advisor_panel.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   advisor_panel.sh

vboxuser@Jomanah:~/project/Software-Tools-Project$ git commit -m "Added advisor_panel.sh script"
[main 2016bf8] Added advisor_panel.sh script
 1 file changed, 41 insertions(+)
 create mode 100644 advisor_panel.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ git config --global user.name "JomanahAhmeds"
vboxuser@Jomanah:~/project/Software-Tools-Project$ git config --global user.email "jomanahahmeds@gmail.com"
vboxuser@Jomanah:~/project/Software-Tools-Project$ git push origin main
```

```
vboxuser@Jomanah:~/project/Software-Tools-Project$ git push origin main
Username for 'https://github.com': JomanahAhmeds
Password for 'https://JomanahAhmeds@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 679 bytes | 48.00 KiB/s, done.
Total 3 (delta 1), reused 1 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/asail-12/Software-Tools-Project.git
   33e6aa6..2016bf8  main -> main
```

- Script Code

```bash
#!/bin/bash
# script runs the Advisor Panel for viewing student result statistics

results_file="results.csv"
# Set the name of the results file

if [[ ! -f $results_file ]]; then
  # Check if the results file does not exist
  echo "No results available. Please run prediction first."
  exit 1
  # Exit the script if file is missing
fi

while true; do
  # Infinite loop to keep showing the menu until the user exits
  echo ""
  echo "===== Advisor Panel =====" 
  echo "1. View all student results"
  echo "2. View major distribution"
  echo "3. View top 3 most selected majors"
  echo "4. View number of students per date"
  echo "5. Exit"
  read -p "Enter your choice (1-5): " choice
  # Display menu and read user input

  case $choice in
    1)
      # Option 1: Show the full results table with aligned columns
      column -s, -t < "$results_file"
      ;;
    2)
      # Option 2: Show how many students selected each major
      cut -d',' -f3 "$results_file" | sort | uniq -c | sort -nr
      ;;
    3)
      # Option 3: Show the top 3 most frequently selected majors
      cut -d',' -f3 "$results_file" | sort | uniq -c | sort -nr | head -n 3
      ::
    4)
      # Option 4: Show the number of students for each date
      cut -d',' -f2 "$results_file" | sort | uniq -c | sort -nr
      ;;
    5)
      # Option 5: Exit the loop and close the panel
      echo "Exiting Advisor Panel."
      break
      ;;
    *)
      # Handle invalid menu choices
      echo "Invalid choice."
      ;;
  esac
done
```

- **The Output**

```
vboxuser@Jomanah:~/project/Software-Tools-Project$ chmod +x advisor_panel.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ ./advisor_panel.sh

===== Advisor Panel =====
1. View all student results
2. View major distribution
3. View top 3 most selected majors
4. View number of students per date
5. Exit
Enter your choice (1-5): 1
asail  2025-05-17  Computer Science

===== Advisor Panel =====
1. View all student results
2. View major distribution
3. View top 3 most selected majors
4. View number of students per date
5. Exit
Enter your choice (1-5): 2
      1 Computer Science
```

- The output shows all student records from the results.csv file in a clear table format. It also displays how many students got each major and identifies the most and least selected majors.

2.4.4 search_tool.sh

This script allows users to search for student records in the results.csv file based on name, predicted major, or date of entry.

- **Terminal Upload Commands**

```
vboxuser@Hajer:~$ mkdir project
vboxuser@Hajer:~$ cd project
vboxuser@Hajer:~/project$ git clone https://github.com/asail-12/Software-Tools-Project.git
Cloning into 'Software-Tools-Project'...
remote: Enumerating objects: 78, done.
remote: Counting objects: 100% (78/78), done.
remote: Compressing objects: 100% (63/63), done.
remote: Total 78 (delta 27), reused 32 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (78/78), 22.76 KiB | 298.00 KiB/s, done.
Resolving deltas: 100% (27/27), done.
vboxuser@Hajer:~/project$ ls
Software-Tools-Project
vboxuser@Hajer:~/project$ cd Software-Tools-Project
vboxuser@Hajer:~/project/Software-Tools-Project$ ls
Interests_Dataset.csv  results.csv  smartmajor_project.sh  smartmajor_stats.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ git add search_tool.sh protect_data.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   protect_data.sh
        new file:   search_tool.sh

vboxuser@Hajer:~/project/Software-Tools-Project$ git commit -m "Added search_tool.sh protect_data.sh"

[main 33e6aa6] Added search_tool.sh protect_data.sh
 2 files changed, 67 insertions(+)
 create mode 100755 protect_data.sh
 create mode 100755 search_tool.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ git config --global user.name "HajerSayah"
vboxuser@Hajer:~/project/Software-Tools-Project$ git config --global user.name "HajerSayah"
vboxuser@Hajer:~/project/Software-Tools-Project$ git config --global user.email"hsayah870@gmail.com"

vboxuser@Hajer:~/project/Software-Tools-Project$ git push origin main
Username for 'https://github.com': HajerSayah
Password for 'https://HajerSayah@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.29 KiB | 1.29 MiB/s, done.
Total 4 (delta 1), reused 2 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/asail-12/Software-Tools-Project.git
   791c7d7..33e6aa6  main -> main
```

- ## Script Code

```bash
#!/bin/bash

result_file="results.csv"
# Name of the results file we are searching

if [[ ! -f $result_file ]]; then
    echo "Results file not found "
    exit 1
fi
# If the file does not exist, we print a message and stop the script.

echo "--- Search Tool ---"
echo "1. Search by student name"
echo "2. Search by predicted major"
echo "3. Search by date (YYYY-MM-DD)"
#Show search options to the user

read -p "Enter your choice 1/2/3 : " option
# We ask the user to choose the type of search.

case $option in
    1) read -p "Enter student name: " name
    # If he chooses 1, he will be asked for the student's name.
       grep -i "$name" "$result_file";;                                    #Search for the name within

    2) read -p "Enter major name: " major
    # If he chooses 2, he is asked for the name of his expected major.
       grep -i "$major" "$result_file";;
    # Looking for a major in the file

    3) read -p "Enter date: " date
    # If he chooses 3, he is asked for the date.
       grep -i "$date" "$result_file";;
    # Looking for the date in the file

    *) echo "Invalid choice";;
    # Any option other than 1, 2 or 3 is considered invalid.

esac
```

- ## The Output

```
vboxuser@Hajer:~/project/Software-Tools-Project$ chmod +x search_tool.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ ./search_tool.sh
--- Search Tool ---
1. Search by student name
2. Search by predicted major
3. Search by date (YYYY-MM-DD)
Enter your choice 1/2/3 : 1
Enter student name: asail
asail,2025-05-17,Computer Science
```

- The output shows the matching student records from the results.csv file based on the search option selected name, major, or date.

2.4.5 protect_data.sh

This script applies file permissions to protect the results.csv file by making it read-only, preventing unauthorized changes.

- **Terminal Upload Commands**

```
vboxuser@Hajer:~$ mkdir project
vboxuser@Hajer:~$ cd project
vboxuser@Hajer:~/project$ git clone https://github.com/asail-12/Software-Tools-Project.git
Cloning into 'Software-Tools-Project'...
remote: Enumerating objects: 78, done.
remote: Counting objects: 100% (78/78), done.
remote: Compressing objects: 100% (63/63), done.
remote: Total 78 (delta 27), reused 32 (delta 9), pack-reused 0 (from 0)
Receiving objects: 100% (78/78), 22.76 KiB | 298.00 KiB/s, done.
Resolving deltas: 100% (27/27), done.
vboxuser@Hajer:~/project$ ls
Software-Tools-Project
vboxuser@Hajer:~/project$ cd Software-Tools-Project
vboxuser@Hajer:~/project/Software-Tools-Project$ ls
Interests_Dataset.csv  results.csv  smartmajor_project.sh  smartmajor_stats.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ git add search_tool.sh protect_data.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   protect_data.sh
        new file:   search_tool.sh

vboxuser@Hajer:~/project/Software-Tools-Project$ git commit -m "Added search_tool.sh protect_data.sh"

[main 33e6aa6] Added search_tool.sh protect_data.sh
 2 files changed, 67 insertions(+)
 create mode 100755 protect_data.sh
 create mode 100755 search_tool.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ git config --global user.name "HajerSayah"
vboxuser@Hajer:~/project/Software-Tools-Project$ git config --global user.name "HajerSayah"
vboxuser@Hajer:~/project/Software-Tools-Project$ git config --global user.email"hsayah870@gmail.com"

vboxuser@Hajer:~/project/Software-Tools-Project$ git push origin main
Username for 'https://github.com': HajerSayah
Password for 'https://HajerSayah@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.29 KiB | 1.29 MiB/s, done.
Total 4 (delta 1), reused 2 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/asail-12/Software-Tools-Project.git
   791c7d7..33e6aa6  main -> main
```

- **Script Code**

```
#!/bin/bash

results_file="results.csv"
# Name of the file we will secure

if [[ ! -f $results_file ]]; then
    echo "Error: '$results_file' does not exist"
    exit 1
fi
# If the file does not exist, we print a message and stop.

read -p "Do you want to secure the results file? yes/no: " answer
# We ask the user if he wants to secure the file.

if [[ "$answer" == "yes" || "$answer" == "y" ]]; then
# If he says yes or y
    chmod 600 "$results_file"
# We give the file read and write permissions only to the file owner
    umask 077
# We set the umask so that any new files are automatically secured (read & write only for the owner)
    echo "File Secured"
# We print a confirmation message
    ls -l "$results_file"
# We show the file permissions after securing
else
    echo "Action Cancelled"
# If the user declines, we print a cancellation message.
fi
```

- **The Output**

```
vboxuser@Hajer:~/project/Software-Tools-Project$ chmod +x protect_data.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ ./protect_data.sh
Do you want to secure the results file? yes/no: no
Action Cancelled
```

- The output shows that the results.csv file has been successfully protected. It is now read-only and cannot be edited or changed, which keeps student data safe.

2.5 justification of design decisions

- We selected a simple text-based design to guarantee that the application stays small and works with a variety of systems without requiring complicated settings or extra applications. Because accessibility is given top priority, users can run the program with ease on any device that has a standard command-line interface.
- The application walks users through the procedure with simple input prompts and logical, unambiguous phases. The system reduces user uncertainty and makes interaction intuitive, especially for people with little technical expertise, by concentrating on straightforward yes/no questions and straightforward data processing.
- The primary goal was to create a useful and intuitive tool that helps students select their university major without adding needless complication. Prioritizing ease of use and functionality guarantees that students may obtain results fast and make wise selections with little effort.

3. Testing and Results

All scripts were tested using the terminal to ensure correct functionality and accurate results. The testing process used the Interests_Dataset.csv file, and outputs were saved in results.csv. The goal was to verify that each script performs its task as expected and handles user input properly.

3.1 smartmajor_predict.sh

- **Testing Strategy:**
  The script was run with different combinations of "yes" and "no" answers to the interest questions. The predicted major was compared to the dataset to check if the match was accurate.
- **Expected Result:**
  The script should recommend the major with the highest matching interests.

```
asail@asail:~/project/Software-Tools-Project$ ./smartmajor_project.sh
Enter your full name:
asail
Please answer with yes or no:
Are you interested in "programming"? yes
Are you interested in "problem_solving"? yes
Are you interested in "biology"? no
Are you interested in "drawing"? no
Are you interested in "debate"? no
Are you interested in "leadership"? yes
Are you interested in "technology"? yes
Are you interested in "healthcare"? no
Are you interested in "math"? no
Are you interested in "reading"? no

The most suitable major for asail is: Computer Science
```

3.2 smartmajor_stats.sh

- **Testing Strategy:**
  After adding multiple records to results.csv, the script was run to calculate how many students were matched with each major.
- **Expected Result:**
  A summary of major counts should be displayed.

```
vboxuser@Jomanah:~/project/Software-Tools-Project$ chmod +x smartmajor_stats.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ ./smartmajor_stats.sh
View and Statistics Menu:
1. View all student results
2. View statistics summary
Enter your choice (1 or 2): 1
Student results:
asail,2025-05-17,Computer Science
vboxuser@Jomanah:~/project/Software-Tools-Project$ ./smartmajor_stats.sh
View and Statistics Menu:
1. View all student results
2. View statistics summary
Enter your choice (1 or 2): 2
Total students: 1
Major predictions
      1 Computer Science
```

3.3 advisor_panel.sh

- **Testing Strategy:**
  The script was run to show all student records and to identify the most and least selected majors.
- **Expected Result:**
  Display of all records in columns, and a summary of most and least common majors.

```
vboxuser@Jomanah:~/project/Software-Tools-Project$ chmod +x advisor_panel.sh
vboxuser@Jomanah:~/project/Software-Tools-Project$ ./advisor_panel.sh

===== Advisor Panel =====
1. View all student results
2. View major distribution
3. View top 3 most selected majors
4. View number of students per date
5. Exit
Enter your choice (1-5): 1
asail  2025-05-17  Computer Science

===== Advisor Panel =====
1. View all student results
2. View major distribution
3. View top 3 most selected majors
4. View number of students per date
5. Exit
Enter your choice (1-5): 2
      1 Computer Science
```

3.4 search_tool.sh

- **Testing Strategy:**
  Searches were performed using student name, predicted major, and date to verify the script's ability to filter the results.
- **Expected Result:**
  Only matching records should be shown from results.csv.

```
vboxuser@Hajer:~/project/Software-Tools-Project$ chmod +x search_tool.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ ./search_tool.sh
--- Search Tool ---
1. Search by student name
2. Search by predicted major
3. Search by date (YYYY-MM-DD)
Enter your choice 1/2/3 : 1
Enter student name: asail
asail,2025-05-17,Computer Science
```

3.5 protect_data.sh

- **Testing Strategy:**
  The script was executed, and then an attempt was made to edit results.csv.
- **Expected Result:**
  The file should become read-only to prevent changes.

```
vboxuser@Hajer:~/project/Software-Tools-Project$ chmod +x protect_data.sh
vboxuser@Hajer:~/project/Software-Tools-Project$ ./protect_data.sh
Do you want to secure the results file? yes/no: no
Action Cancelled
```

- All scripts worked as expected and gave the correct results. This confirmed that SmartMajor is reliable and meets its goals.

4. Conclusion

4.1 Summary of achievements

The SmartMajor project is a complete and simple system that helps students choose the most suitable university major based on their interests. The project idea was clear from the beginning, and the work was divided between the team members across five Bash scripts. Each script had a specific role: predicting majors, showing statistics, searching results, managing advisor tools, and protecting student data.
SmartMajor solves a real problem that many students face when trying to decide their future path. Instead of guessing or following others, students can answer a few yes/no questions, and the system compares their answers with a dataset of majors and interests. In the end, they get a helpful and accurate suggestion that fits their personality and goals.

4.2 Limitations

Limited dataset: It needs more diverse interests and specializations to reflect the evolving educational landscape.

Text-based user interface: Although simple, it may confuse some students and reduce the tool's appeal.

Difficulty uploading the project to GitHub: We had some problems while using Git and trying to upload the scripts, such as permission errors, large file sizes, and setting up SSH keys. These issues caused delays in sharing the code with others.

4.3 potential future work.

Updating the dataset regularly with new majors and interests based on changes in education and the job market. We can also allow advisors to add new data through an improved dashboard.
Creating a graphical interface (using tools like GTK or Electron) to make the system easier to use. It can also include support for different languages to help more students benefit from it.
Using machine learning to improve predictions. Instead of only using yes/no answers, the system can learn from previous students' choices to give smarter and more personalized results.